VANISHING OF THE SUM OF THE ORDERS OF THE TAME SYMBOLS OVER THE HEIGHT ONE PRIMES OF A TWO-DIMENSIONAL NOETHERIAN LOCAL RING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Preparation of the tame symbol along each height one prime. Same chapter chow.tex, labels lemma-prepare-tame-symbol (proof line 1153) and lemma-perpare-key (proof line 1154), the latter applied with t = ab. Produces, for each height one prime q_i, a finite local extension A inside B with B/A killed by a power of ab and B_{q_i} = B_i; passing to B is the first reduction and is carried out in the counted span.
- Norm compatibility of the tame symbol along a finite extension. Same chapter chow.tex, label lemma-norm-down-tame-symbol (proof line 1166); Algebra, label algebra-lemma-finite-extension-dim-1 (proof line 1167). Expresses the order of the tame symbol at q_i as the sum of the orders at the primes of B over q_i, so that the replacement of A by B preserves the total sum.
- Comparison of periodic lengths and the multiplication formula. Same chapter chow.tex, labels lemma-compare-periodic-lengths (proof line 1195), lemma-multiply-period-length (proof line 1218), lemma-powers-period-length-zero (proof line 1225) and lemma-length-multiplication (proof lines 1227, 1241, 1257). Turns the vanishing of the sum of the e_A(M_i, a, b) into the statement about orders, and supplies the term by term identities used in the computation; the computation itself is inside the counted span.

Source reference. Original lemma 02MJ: sources/stacks-chow/tags.txt, source lines 3446–3446. Navigation pointer for \ref{algebra-lemma-finite-extension-dim-1} used inside the counted span (line 1167); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0EA9: sources/stacks-chow/pinned-original-source.tex, source lines 283–283. Navigation pointer for \ref{lemma-compare-periodic-lengths} used inside the counted span (line 1195); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QF: sources/stacks-chow/pinned-original-source.tex, source lines 310–310. Navigation pointer for \ref{lemma-length-multiplication} used inside the counted span (line 1227); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAC: sources/stacks-chow/pinned-original-source.tex, source lines 464–464. Navigation pointer for \ref{lemma-multiply-period-length} used inside the counted span (line 1218); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAT: sources/stacks-chow/pinned-original-source.tex, source lines 952–952. Navigation pointer for \ref{lemma-norm-down-tame-symbol} used inside the counted span (line 1166); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAV: sources/stacks-chow/pinned-original-source.tex, source lines 1105–1105. Navigation pointer for \ref{lemma-perpare-key} used inside the counted span (line 1154); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAB: sources/stacks-chow/pinned-original-source.tex, source lines 404–404. Navigation pointer for \ref{lemma-powers-period-length-zero} used inside the counted span (line 1225); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAG: sources/stacks-chow/pinned-original-source.tex, source lines 642–642. Navigation pointer for \ref{lemma-prepare-tame-symbol} used inside the counted span (line 1153); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a 2-dimensional Noetherian local ring. Let $a, b \in A$ be nonzerodivisors. Then we have*

$$\sum ord_{A/\mathfrak{q}}(\partial_{A_\mathfrak{q}}(a, b)) = 0$$

where the sum is over the height 1 primes $\mathfrak{q}$ of A.

Proof. If $\mathfrak{q}$ is a height 1 prime of A such that a, b map to a unit of $A_\mathfrak{q}$, then $\partial_{A_\mathfrak{q}}(a, b) = 1$. Thus the sum is finite. In fact, if $V(ab) = \{\mathfrak{m}, \mathfrak{q}_1, \ldots, \mathfrak{q}_r\}$ then the sum is over $i = 1, \ldots, r$. For each i we pick an extension $A_{\mathfrak{q}_i} \subset B_i$ as in Lemma 0EAG for a, b. By Lemma 0EAV with $t = ab$ and the given list of primes we may assume we have a finite local extension $A \subset B$ with B/A annihilated by a power of ab and such that for each i the $B_{\mathfrak{q}_i} \cong B_i$. Observe that if $\mathfrak{q}_{i,j}$ are the primes of B lying over $\mathfrak{q}_i$ then we have

$$\mathrm{ord}_{A/\mathfrak{q}_i}(\partial_{A_{\mathfrak{q}_i}}(a, b)) = \sum\nolimits_j \mathrm{ord}_{B/\mathfrak{q}_{i,j}}(\partial_{B_{\mathfrak{q}_{i,j}}}(a, b))$$

by Lemma 0EAT and Algebra, Lemma 02MJ. Thus we may replace A by B and reduce to the case discussed in the next paragraph.

Assume for each i there is a nonzerodivisor $\pi_i \in A_{\mathfrak{q}_i}$ and units $u_i, v_i \in A_{\mathfrak{q}_i}$ such that for some integers $e_i, f_i \geq 0$ we have

$$a = u_i \pi_i^{e_i}, \quad b = v_i \pi_i^{f_i}$$

in $A_{\mathfrak{q}_i}$. Setting $m_i = \text{length}_{A_{\mathfrak{q}_i}}(A_{\mathfrak{q}_i}/\pi_i)$ we have $\partial_{A_{\mathfrak{q}_i}}(a,b) = ((-1)^{e_i f_i} u_i^{f_i} v_i^{-e_i})^{m_i}$ by definition. Since a, b are nonzerodivisors the $(2,1)$-periodic complex $(A/(ab), a, b)$ has vanishing cohomology. Denote M_i the image of $A/(ab)$ in $A_{\mathfrak{q}_i}/(ab)$. Then we have a map

$$A/(ab) \longrightarrow \bigoplus M_i$$

whose kernel and cokernel are supported in $\{\mathfrak{m}\}$ and hence have finite length. Thus we see that

$$\sum e_A(M_i, a, b) = 0$$

by Lemma 0EA9. Hence it suffices to show $e_A(M_i, a, b) = -\text{ord}_{A/\mathfrak{q}_i}(\partial_{A_{\mathfrak{q}_i}}(a,b))$.

Let us prove this first, in case π_i, u_i, v_i are the images of elements $\pi_i, u_i, v_i \in A$ (using the same symbols should not cause any confusion). In this case we get

$$\begin{aligned} e_A(M_i, a, b) &= e_A(M_i, u_i\pi_i^{e_i}, v_i\pi_i^{f_i}) \\ &= e_A(M_i, \pi_i^{e_i}, \pi_i^{f_i}) - e_A(\pi_i^{e_i}M_i, 0, u_i) + e_A(\pi_i^{f_i}M_i, 0, v_i) \\ &= 0 - f_i m_i \text{ord}_{A/\mathfrak{q}_i}(u_i) + e_i m_i \text{ord}_{A/\mathfrak{q}_i}(v_i) \\ &= -m_i \text{ord}_{A/\mathfrak{q}_i}(u_i^{f_i} v_i^{-e_i}) = -\text{ord}_{A/\mathfrak{q}_i}(\partial_{A_{\mathfrak{q}_i}}(a,b)) \end{aligned}$$

The second equality holds by Lemma 0EAC. Observe that $M_i \subset (M_i)_{\mathfrak{q}_i} = A_{\mathfrak{q}_i}/(\pi_i^{e_i+f_i})$ and $(\pi_i^{e_i}M_i)_{\mathfrak{q}_i} \cong A_{\mathfrak{q}_i}/\pi_i^{f_i}$ and $(\pi_i^{f_i}M_i)_{\mathfrak{q}_i} \cong A_{\mathfrak{q}_i}/\pi_i^{e_i}$. The 0 in the third equality comes from Lemma 0EAB and the other two terms come from Lemma 02QF. The last two equalities follow from multiplicativity of the order function and from the definition of our tame symbol.

In general, we may first choose $c \in A$, $c \notin \mathfrak{q}_i$ such that $c\pi_i \in A$. After replacing π_i by $c\pi_i$ and u_i by $c^{-e_i}u_i$ and v_i by $c^{-f_i}v_i$ we may and do assume π_i is in A. Next, choose an $c \in A$, $c \notin \mathfrak{q}_i$ with $cu_i, cv_i \in A$. Then we observe that

$$e_A(M_i, ca, cb) = e_A(M_i, a, b) - e_A(aM_i, 0, c) + e_A(bM_i, 0, c)$$

by Lemma 02QF. On the other hand, we have

$$\partial_{A_{\mathfrak{q}_i}}(ca, cb) = c^{m_i(f_i - e_i)}\partial_{A_{\mathfrak{q}_i}}(a,b)$$

in $\kappa(\mathfrak{q}_i)^*$ because c is a unit in $A_{\mathfrak{q}_i}$. The arguments in the previous paragraph show that $e_A(M_i, ca, cb) = -\text{ord}_{A/\mathfrak{q}_i}(\partial_{A_{\mathfrak{q}_i}}(ca, cb))$. Thus it suffices to prove

$$e_A(aM_i, 0, c) = \text{ord}_{A/\mathfrak{q}_i}(c^{m_i f_i}) \quad \text{and} \quad e_A(bM_i, 0, c) = \text{ord}_{A/\mathfrak{q}_i}(c^{m_i e_i})$$

and this follows from Lemma 02QF by the description (see above) of what happens when we localize at $\mathfrak{q}_i$. $\square$